Config.php

```php
<?php
declare(strict_types=1);

session_start([
   'cookie_httponly' => true,
   'cookie_samesite' => 'Lax',
   'use_strict_mode' => true,
]);

const DB_HOST = '127.0.0.1';
const DB_NAME = 'home_crud';
const DB_USER = 'root';
const DB_PASS = '';

const REMEMBER_COOKIE_EXPIRES = 2592000; // 30 days

function db(): PDO
{
   static $pdo = null;

   if ($pdo instanceof PDO) {
       return $pdo;
   }

   $dsn = sprintf('mysql:host=%s;dbname=%s;charset=utf8mb4', DB_HOST, DB_NAME);

   $pdo = new PDO($dsn, DB_USER, DB_PASS, [
       PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
       PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
       PDO::ATTR_EMULATE_PREPARES => false,
   ]);

   return $pdo;
}

function e(?string $value): string
{
   return htmlspecialchars((string) $value, ENT_QUOTES, 'UTF-8');
}

function redirect(string $path): never
```

```
{
   header('Location: ' . $path);
   exit;
}

function set_flash(string $type, string $message): void
{
   $_SESSION['flash'] = [
       'type' => $type,
       'message' => $message,
   ];
}

function get_flash(): ?array
{
   $flash = $_SESSION['flash'] ?? null;
   unset($_SESSION['flash']);

   return $flash;
}

function csrf_token(): string
{
   if (empty($_SESSION['csrf_token'])) {
       $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
   }

   return $_SESSION['csrf_token'];
}

function verify_csrf(): void
{
   $token = $_POST['csrf_token'] ?? '';

   if (!is_string($token) || !hash_equals($_SESSION['csrf_token'] ?? '', $token)) {
       http_response_code(419);
       exit('Invalid CSRF token.');
   }
}

function normalize_input(array $source): array
{
```

```php
    return [
        'name' => trim((string) ($source['name'] ?? '')),
        'email' => trim((string) ($source['email'] ?? '')),
        'phone' => trim((string) ($source['phone'] ?? '')),
        'city' => trim((string) ($source['city'] ?? '')),
    ];
}

function validate_user(array $input): array
{
    $errors = [];

    if ($input['name'] === '') {
        $errors['name'] = 'Name is required.';
    } elseif (mb_strlen($input['name']) > 100) {
        $errors['name'] = 'Name must be 100 characters or fewer.';
    }

    if ($input['email'] === '') {
        $errors['email'] = 'Email is required.';
    } elseif (!filter_var($input['email'], FILTER_VALIDATE_EMAIL)) {
        $errors['email'] = 'Please enter a valid email address.';
    } elseif (mb_strlen($input['email']) > 150) {
        $errors['email'] = 'Email must be 150 characters or fewer.';
    }

    if ($input['phone'] !== '' && !preg_match('/^[0-9+\-\s]{7,20}$/', $input['phone']))
{
        $errors['phone'] = 'Phone should contain only numbers, spaces, + or - signs.';
    }

    if ($input['city'] !== '' && mb_strlen($input['city']) > 100) {
        $errors['city'] = 'City must be 100 characters or fewer.';
    }

    return $errors;
}

function remember_user_input(array $input): void
{
    $expires = time() + REMEMBER_COOKIE_EXPIRES;
```

```php
    setcookie('remember_name', $input['name'], [
        'expires' => $expires,
        'path' => '/',
        'httponly' => true,
        'samesite' => 'Lax',
    ]);

    setcookie('remember_email', $input['email'], [
        'expires' => $expires,
        'path' => '/',
        'httponly' => true,
        'samesite' => 'Lax',
    ]);
}

function set_last_visit_cookie(): void
{
    setcookie('last_visit', date('Y-m-d H:i:s'), [
        'expires' => time() + REMEMBER_COOKIE_EXPIRES,
        'path' => '/',
        'httponly' => true,
        'samesite' => 'Lax',
    ]);
}

function previous_input(): array
{
    return $_SESSION['old_input'] ?? [];
}

function previous_errors(): array
{
    return $_SESSION['errors'] ?? [];
}

function flash_form_state(array $input, array $errors): void
{
    $_SESSION['old_input'] = $input;
    $_SESSION['errors'] = $errors;
}

function clear_form_state(): void
```

```
{
   unset($_SESSION['old_input'], $_SESSION['errors']);
}
```

Create.php

```
<?php
declare(strict_types=1);

require __DIR__ . '/config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
   redirect('/Home/project/index.php');
}

verify_csrf();

$input = normalize_input($_POST);
$errors = validate_user($input);

if ($errors) {
   flash_form_state($input, $errors);
   set_flash('error', 'Please fix the highlighted fields and try again.');
   redirect('/Home/project/index.php');
}

try {
   $statement = db()->prepare(
       'INSERT INTO users (name, email, phone, city) VALUES (:name, :email, :phone,
:city)'
   );
   $statement->execute($input);

   remember_user_input($input);
   set_flash('success', 'User created successfully.');
} catch (PDOException $exception) {
   flash_form_state($input, []);
   set_flash('error', 'Unable to save the user. The email may already exist.');
}

redirect('/Home/project/index.php');
```

Delect.php

```php
<?php
declare(strict_types=1);

require __DIR__ . '/config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
   redirect('/Home/project/index.php');
}

verify_csrf();

$id = filter_input(INPUT_POST, 'id', FILTER_VALIDATE_INT);

if (!$id) {
   set_flash('error', 'Invalid user selected for deletion.');
   redirect('/Home/project/index.php');
}

try {
   $statement = db()->prepare('DELETE FROM users WHERE id = :id');
   $statement->execute(['id' => $id]);
   set_flash('success', 'User deleted successfully.');
} catch (PDOException $exception) {
   set_flash('error', 'Unable to delete the user right now.');
}

redirect('/Home/project/index.php');
```

Index.php

```php
<?php
declare(strict_types=1);

require __DIR__ . '/config.php';

$_SESSION['dashboard_visits'] = (int) ($_SESSION['dashboard_visits'] ?? 0) + 1;
```

```
$flash = get_flash();
$formData = previous_input();
$errors = previous_errors();
clear_form_state();

$editId = filter_input(INPUT_GET, 'edit', FILTER_VALIDATE_INT);
$editUser = null;
$dbError = null;
$users = [];

try {
   $pdo = db();

   if ($editId) {
       $statement = $pdo->prepare('SELECT id, name, email, phone, city, created_at,
updated_at FROM users WHERE id = :id');
       $statement->execute(['id' => $editId]);
       $editUser = $statement->fetch();

       if (!$editUser) {
           set_flash('error', 'The selected user does not exist.');
           redirect('/Home/project/index.php');
       }
   }

   $users = $pdo->query('SELECT id, name, email, phone, city, created_at, updated_at
FROM users ORDER BY id DESC')->fetchAll();
} catch (PDOException $exception) {
   $dbError = 'Database connection failed. Create the database from schema.sql and
check config.php credentials.';
}

$isEditing = is_array($editUser);
$prefill = [
   'name' => $formData['name'] ?? ($editUser['name'] ?? ($_COOKIE['remember_name'] ??
'')),
   'email' => $formData['email'] ?? ($editUser['email'] ?? ($_COOKIE['remember_email']
?? '')),
   'phone' => $formData['phone'] ?? ($editUser['phone'] ?? ''),
   'city' => $formData['city'] ?? ($editUser['city'] ?? ''),
];
```

```
$previousVisit = $_COOKIE['last_visit'] ?? 'This is your first visit.';
set_last_visit_cookie();
?>
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>PHP MySQL User Manager</title>
   <link rel="stylesheet" href="/Home/project/styles.css">
</head>
<body>
<main class="page">
   <section class="hero">
       <div>
           <p class="eyebrow">PHP, MySQL, Sessions, Cookies</p>
           <h1>User Management Dashboard</h1>
           <p class="hero-copy">
               Create, review, update, and delete users with server-side validation,
               prepared statements, session flash messages, CSRF protection, and
cookie-based form memory.
           </p>
       </div>
       <div class="hero-stats">
           <div class="stat-card">
               <span>Session visits</span>
               <strong><?= (int) $_SESSION['dashboard_visits'] ?></strong>
           </div>
           <div class="stat-card">
               <span>Last visit</span>
               <strong><?= e($previousVisit) ?></strong>
           </div>
       </div>
   </section>

   <?php if ($flash): ?>
       <div class="alert alert-<?= e($flash['type']) ?>"><?= e($flash['message'])
?></div>
   <?php endif; ?>

   <?php if ($dbError): ?>
       <div class="alert alert-error"><?= e($dbError) ?></div>
```

```
    <?php endif; ?>

    <section class="grid">
        <article class="panel">
            <div class="panel-header">
                <h2><?= $isEditing ? 'Update User' : 'Add User' ?></h2>
                <?php if ($isEditing): ?>
                    <a class="link-button" href="/Home/project/index.php">Cancel
Edit</a>
                <?php endif; ?>
            </div>

            <form method="post" action="<?= $isEditing ? '/Home/project/update.php' :
'/Home/project/create.php' ?>" novalidate>
                <input type="hidden" name="csrf_token" value="<?= e(csrf_token()) ?>">
                <?php if ($isEditing): ?>
                    <input type="hidden" name="id" value="<?= (int) $editUser['id']
?>">
                <?php endif; ?>

                <label for="name">Full Name</label>
                <input id="name" name="name" type="text" maxlength="100" value="<?=
e($prefill['name']) ?>" required>
                <?php if (isset($errors['name'])): ?>
                    <small class="error-text"><?= e($errors['name']) ?></small>
                <?php endif; ?>

                <label for="email">Email Address</label>
                <input id="email" name="email" type="email" maxlength="150" value="<?=
e($prefill['email']) ?>" required>
                <?php if (isset($errors['email'])): ?>
                    <small class="error-text"><?= e($errors['email']) ?></small>
                <?php endif; ?>

                <label for="phone">Phone Number</label>
                <input id="phone" name="phone" type="text" maxlength="20" value="<?=
e($prefill['phone']) ?>" placeholder="+91 9876543210">
                <?php if (isset($errors['phone'])): ?>
                    <small class="error-text"><?= e($errors['phone']) ?></small>
                <?php endif; ?>

                <label for="city">City</label>
```

```
                <input id="city" name="city" type="text" maxlength="100" value="<?=
e($prefill['city']) ?>" placeholder="Ahmedabad">
                <?php if (isset($errors['city'])): ?>
                    <small class="error-text"><?= e($errors['city']) ?></small>
                <?php endif; ?>

                <button class="primary-button" type="submit"><?= $isEditing ? 'Update
User' : 'Save User' ?></button>
            </form>
        </article>

        <article class="panel">
            <div class="panel-header">
                <h2>Stored Users</h2>
                <span class="badge"><?= count($users) ?> records</span>
            </div>

            <div class="table-wrap">
                <table>
                    <thead>
                    <tr>
                        <th>ID</th>
                        <th>Name</th>
                        <th>Email</th>
                        <th>Phone</th>
                        <th>City</th>
                        <th>Created</th>
                        <th>Actions</th>
                    </tr>
                    </thead>
                    <tbody>
                    <?php if (!$dbError && $users): ?>
                        <?php foreach ($users as $user): ?>
                            <tr>
                                <td><?= (int) $user['id'] ?></td>
                                <td><?= e($user['name']) ?></td>
                                <td><?= e($user['email']) ?></td>
                                <td><?= e($user['phone']) ?></td>
                                <td><?= e($user['city']) ?></td>
                                <td><?= e((string) $user['created_at']) ?></td>
                                <td class="actions">
```

```
                                    <a class="link-button"
href="/Home/project/index.php?edit=<?= (int) $user['id'] ?>">Edit</a>
                                    <form method="post"
action="/Home/project/delete.php" onsubmit="return confirm('Delete this user?');">
                                        <input type="hidden" name="csrf_token"
value="<?= e(csrf_token()) ?>">
                                        <input type="hidden" name="id" value="<?= (int)
$user['id'] ?>">
                                        <button class="danger-button"
type="submit">Delete</button>
                                    </form>
                                </td>
                            </tr>
                        <?php endforeach; ?>
                    <?php elseif ($dbError): ?>
                        <tr>
                            <td colspan="7" class="empty-state">The user list is
unavailable until the database connection is fixed.</td>
                        </tr>
                    <?php else: ?>
                        <tr>
                            <td colspan="7" class="empty-state">No users found yet. Add
one using the form.</td>
                        </tr>
                    <?php endif; ?>
                    </tbody>
                </table>
            </div>
        </article>
    </section>
</main>
</body>
</html>
```

SQl

```
CREATE DATABASE IF NOT EXISTS home_crud CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;
USE home_crud;

CREATE TABLE IF NOT EXISTS users (
   id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
```

```
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    phone VARCHAR(20) DEFAULT NULL,
    city VARCHAR(100) DEFAULT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

Style.css

```
:root {
    --bg: #f5efe6;
    --panel: rgba(255, 255, 255, 0.92);
    --panel-border: rgba(37, 33, 31, 0.08);
    --text: #1f1a17;
    --muted: #6d5c52;
    --primary: #b85042;
    --primary-dark: #8e3d31;
    --danger: #b42318;
    --success: #166534;
    --error: #991b1b;
    --shadow: 0 24px 60px rgba(76, 53, 40, 0.14);
}

* {
    box-sizing: border-box;
}

body {
    margin: 0;
    min-height: 100vh;
    font-family: "Trebuchet MS", "Segoe UI", sans-serif;
    color: var(--text);
    background:
        radial-gradient(circle at top left, rgba(184, 80, 66, 0.18), transparent 30%),
        radial-gradient(circle at bottom right, rgba(31, 26, 23, 0.15), transparent
25%),
        linear-gradient(135deg, #f8f3ee, #eadfce 55%, #d9c5af);
}

.page {
```

```
  width: min(1180px, calc(100% - 32px));
  margin: 0 auto;
  padding: 32px 0 48px;
}

.hero {
  display: flex;
  justify-content: space-between;
  gap: 24px;
  align-items: flex-start;
  margin-bottom: 24px;
}

.eyebrow {
  margin: 0 0 8px;
  color: var(--primary);
  font-size: 0.9rem;
  font-weight: 700;
  letter-spacing: 0.08em;
  text-transform: uppercase;
}

h1,
h2 {
  margin: 0;
}

.hero-copy {
  max-width: 680px;
  color: var(--muted);
  line-height: 1.6;
}

.hero-stats {
  display: grid;
  gap: 12px;
  min-width: 240px;
}

.stat-card,
.panel {
  background: var(--panel);
```

```
  border: 1px solid var(--panel-border);
  border-radius: 22px;
  box-shadow: var(--shadow);
  backdrop-filter: blur(8px);
}

.stat-card {
  padding: 18px;
}

.stat-card span {
  display: block;
  color: var(--muted);
  font-size: 0.88rem;
  margin-bottom: 8px;
}

.stat-card strong {
  display: block;
  font-size: 1rem;
  line-height: 1.4;
}

.grid {
  display: grid;
  grid-template-columns: 360px 1fr;
  gap: 24px;
}

.panel {
  padding: 24px;
}

.panel-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 12px;
  margin-bottom: 20px;
}

form {
```

```
  display: grid;
  gap: 12px;
}

label {
  font-weight: 700;
  font-size: 0.95rem;
}

input {
  width: 100%;
  border: 1px solid rgba(31, 26, 23, 0.12);
  border-radius: 14px;
  padding: 12px 14px;
  font-size: 0.98rem;
  transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

input:focus {
  outline: none;
  border-color: rgba(184, 80, 66, 0.8);
  box-shadow: 0 0 0 4px rgba(184, 80, 66, 0.12);
}

.primary-button,
.danger-button,
.link-button {
  border: 0;
  border-radius: 999px;
  padding: 11px 16px;
  font-weight: 700;
  cursor: pointer;
  text-decoration: none;
  text-align: center;
  transition: transform 0.15s ease, opacity 0.15s ease, background 0.15s ease;
}

.primary-button:hover,
.danger-button:hover,
.link-button:hover {
  transform: translateY(-1px);
}
```

```
.primary-button {
   background: var(--primary);
   color: #fff;
}

.primary-button:hover {
   background: var(--primary-dark);
}

.danger-button {
   background: #fee2e2;
   color: var(--danger);
}

.link-button {
   background: rgba(184, 80, 66, 0.12);
   color: var(--primary-dark);
}

.alert {
   margin-bottom: 18px;
   padding: 14px 16px;
   border-radius: 16px;
   font-weight: 700;
}

.alert-success {
   background: rgba(22, 101, 52, 0.12);
   color: var(--success);
}

.alert-error {
   background: rgba(153, 27, 27, 0.1);
   color: var(--error);
}

.error-text {
   color: var(--error);
   margin-top: -4px;
}
```

```
.badge {
   display: inline-flex;
   align-items: center;
   justify-content: center;
   min-width: 96px;
   padding: 6px 12px;
   border-radius: 999px;
   background: rgba(31, 26, 23, 0.08);
   color: var(--muted);
   font-size: 0.88rem;
   font-weight: 700;
}

.table-wrap {
   overflow-x: auto;
}

table {
   width: 100%;
   border-collapse: collapse;
}

th,
td {
   text-align: left;
   padding: 14px 10px;
   border-bottom: 1px solid rgba(31, 26, 23, 0.08);
   vertical-align: top;
}

th {
   color: var(--muted);
   font-size: 0.88rem;
   text-transform: uppercase;
   letter-spacing: 0.04em;
}

.actions {
   display: flex;
   gap: 10px;
   align-items: center;
   flex-wrap: wrap;
```

```
}

.actions form {
   display: inline-flex;
}

.empty-state {
   text-align: center;
   color: var(--muted);
   padding: 36px 12px;
}

@media (max-width: 920px) {
   .hero,
   .grid {
       grid-template-columns: 1fr;
       display: grid;
   }
}

@media (max-width: 640px) {
   .page {
       width: min(100% - 20px, 100%);
       padding-top: 20px;
   }

   .panel,
   .stat-card {
       border-radius: 18px;
       padding: 18px;
   }

   th,
   td {
       padding-inline: 6px;
   }
}
```

Update.php

```
<?php
declare(strict_types=1);
```

```php
require __DIR__ . '/config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
   redirect('/Home/project/index.php');
}

verify_csrf();

$id = filter_input(INPUT_POST, 'id', FILTER_VALIDATE_INT);
$input = normalize_input($_POST);
$errors = validate_user($input);

if (!$id) {
   set_flash('error', 'Invalid user selected for update.');
   redirect('/Home/project/index.php');
}

if ($errors) {
   flash_form_state($input, $errors);
   set_flash('error', 'Please fix the highlighted fields before updating.');
   redirect('/Home/project/index.php?edit=' . $id);
}

try {
   $statement = db()->prepare(
       'UPDATE users
        SET name = :name, email = :email, phone = :phone, city = :city
        WHERE id = :id'
   );
   $statement->execute($input + ['id' => $id]);

   remember_user_input($input);
   set_flash('success', 'User updated successfully.');
} catch (PDOException $exception) {
   flash_form_state($input, []);
   set_flash('error', 'Unable to update the user. The email may already exist.');
   redirect('/Home/project/index.php?edit=' . $id);
}

redirect('/Home/project/index.php');
```

PHP, MYSQL, SESSIONS, COOKIES

User Management Dashboard

Create, review, update, and delete users with server-side validation, prepared statements, session flash messages, CSRF protection, and cookie-based form memory.

Session visits
4

Last visit
2026-04-06 12:02:37

Add User

Full Name
RAHUL

Email Address
maguvof0jed2@maximail.vip

Phone Number
+91 9876543210

City
Ahmedabad

Save User

Stored Users

1 records

ID	NAME	EMAIL	PHONE	CITY	CREATED	ACTIONS
1	RAHUL	maguvof0jed2@maximail.vip	+91 87349 73923		2026-04-06 15:32:37	Edit Delete